# TIP SERIES

## PROFESSIONAL POWER AMPLIFIER

| TIP 1002 | TIP 1302 | TIP 1602 | TIP 2402 |
|---|---|---|---|
| 1000W × 2 @ 8Ω | 1300W × 2 @ 8Ω | 1600W × 2 @ 8Ω | 2400W × 2 @ 8Ω |
| 3000W Bridge | 3800W Bridge | 4800W Bridge | 8000W Bridge |

| | |
|---|---|
| *Document Type* | User Manual / Installation Guide |
| *Revision* | Rev. 1.0 |
| *Models Covered* | TIP 1002 \| TIP 1302 \| TIP 1602 \| TIP 2402 |
| *Manufacturer* | Louis Martin Audio \| www.louismartinaudio.it |

## Table of Contents

## 1. Safety Instructions

- Read all instructions carefully before operating.
- Do not use this product near water or in damp environments.
- Clean only with a dry cloth. Never use liquid cleaners.
- Do not block ventilation openings. Maintain at least 1U (44mm) clearance above and below in racks.
- Do not install near heat sources (radiators, stoves, other amplifiers).
- Protect the power cord from pinching, especially at plugs and the unit entry point.
- Unplug during lightning storms or extended periods of non-use.
- All servicing must be carried out by qualified service personnel only.
- Ensure the mains supply voltage matches the amplifier's rated voltage (110–240V AC, 50/60Hz).
- Disconnect power before making or changing speaker connections.

***NOTE:** This unit contains no user-serviceable parts inside. Opening the chassis voids the warranty and may expose lethal voltages.*

## 2. Product Overview

The TIP Series is a range of professional multi-channel Class-D power amplifiers engineered by Louis Martin Audio for demanding live sound, installation, and touring applications. Combining high power output with lightweight construction, intelligent thermal management, and comprehensive protection circuitry, the TIP Series delivers exceptional audio fidelity with outstanding reliability.

### Key Features:

- Class-D switching amplifier topology for maximum efficiency (>85%)
- Automatic temperature-controlled variable-speed fan cooling
- Multi-mode operation: Stereo, Parallel, and Bridge Mono
- Full protection: short-circuit, DC offset, over-temperature, over-current, clip limiting
- Balanced XLR inputs with 20kΩ impedance for noise-free long cable runs
- Neutrik Speakon speaker outputs rated for professional loudspeaker systems
- Signal, Clip and Protect LED indicators per channel
- 19-inch rack-mountable chassis (2U, 88mm height)
- Universal switch-mode power supply: 110–240V AC, 50/60Hz auto-sensing

## 3. Package Contents

**1 ×** TIP Series Power Amplifier (specified model)
**1 ×** IEC AC Power Cord (country-appropriate)
**4 ×** Rack Mounting Screws (M6, with plastic washers)
**1 ×** User Manual (this document)
**1 ×** Warranty Registration Card

***NOTE:** Inspect the packaging for shipping damage before use. Contact your dealer immediately if any items are missing or damaged.*

## 4. Front Panel Controls


Figure 1 — TIP Series Front Panel

| | |
|---|---|
| **Ventilation Slots** | Perforated grille providing airflow for the internal heat sink and fan. Never obstruct. |
| **Channel A / B Gain Knob** | Stepped rotary attenuator (0–10 scale). Controls individual channel output level. Fully clockwise = maximum gain. |
| **Signal LED (Green)** | Illuminates when audio signal is present above –40 dBu. Confirms active signal path. |
| **Clip LED (Orange)** | Illuminates when the output approaches clipping (approx. –3 dBFS). Reduce source level or amp gain. |
| **Protect LED (Red)** | Illuminates during a protection event: over-temperature, DC fault, short-circuit, or over-current. Unit mutes automatically. |
| **Power Switch** | Rocker switch (ON / OFF). Always power amplifiers on last in a signal chain and off first to prevent transient pops. |

## 5. Rear Panel Connections


Figure 2 — TIP Series Rear Panel

| | |
|---|---|
| **Cooling Fins / Fan Exhaust** | Always ensure unobstructed airflow. Minimum 100mm clearance behind unit when rack-mounted. |
| **AC Mains IEC Socket** | Connect supplied power cord. The unit accepts 110–240V AC, 50/60Hz (auto-sensing). |
| **XLR Input CH A / CH B** | Balanced XLR-3F connectors. Pin 1: GND, Pin 2: Hot (+), Pin 3: Cold (–). Accept signal levels from 0.775V to 1.44V (+4 dBu). |
| **Speakon Output CH A / CH B** | 4-pole Neutrik Speakon connectors. Pins 1+ / 1– carry speaker signal. Use rated Speakon cables only. |
| **Sub Input / Sub Output** | Loop-through connector for sub-bass distribution in multi-amp systems. |
| **Bridge / Parallel Switch** | Selects amplifier operating mode. See Section 10 for full details. |
| **Stereo / Bridge Toggle** | Alternate labeling on some variants. Always match wiring to selected mode. |
| **Chassis Ground Screw** | Earth bonding point. Connect to rack chassis ground for hum elimination in large installations. |

## 6. Technical Specifications

| Specification | TIP 1002 | TIP 1302 | TIP 1602 | TIP 2402 |
|---|---|---|---|---|
| Power @ 8Ω (per ch) | 1000W × 2 | 1300W × 2 | 1600W × 2 | 2400W × 2 |
| Power @ 4Ω (per ch) | 1600W × 2 | 2000W × 2 | 2600W × 2 | 4000W × 2 |
| Bridge @ 8Ω | 3000W | 3800W | 4800W | 8000W |
| Avg. Current Draw | 5A / 230V | 6A / 230V | 7A / 230V | 8A / 230V |
| Max Power Consumption | 1150W | 1380W | 1610W | 1840W |
| Frequency Response | 20Hz – 20kHz (±0.5 dB) | | | |
| THD+N | <0.05% @ 8Ω, 1kHz | | | |
| S/N Ratio | >98 dB (A-weighted) | | | |
| Damping Factor | >800 @ 1kHz/8Ω | | | |
| Crosstalk | >85 dB @ 1kHz | | | |
| Input Sensitivity | 0.775V / 1.44V / 32 dB | | | |
| Input Impedance | 20kΩ Balanced | | | |
| Dimensions (W×D×H) | 482 × 360 × 88 mm | | | |
| Net Weight | 5.8 kg | 6.5 kg | 6.8 kg | 7 kg |

***NOTE:*** *All power ratings are measured at rated THD+N with both channels driven simultaneously. Bridge power ratings use a single 8Ω load.*

## 7. Installation & Rack Mounting

### Rack Mounting


Figure 3 — TIP Series in a Standard 19" Rack (2U)

The TIP Series amplifiers are designed for standard 19-inch equipment racks and occupy 2U (88mm) of rack space. Follow these steps for safe installation:

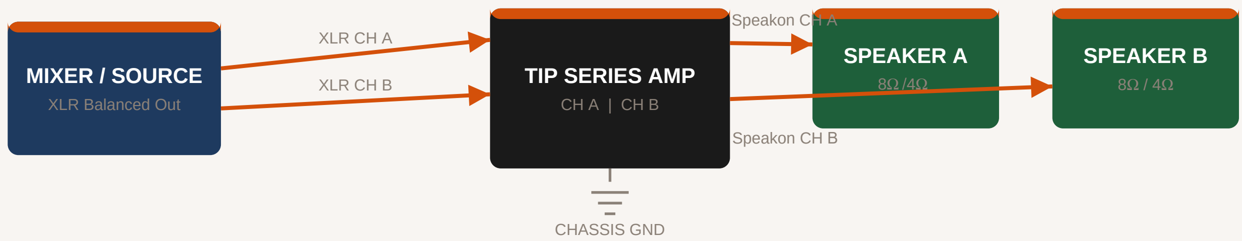
1. Ensure the rack is stable and level before installation.
2. Use all four supplied M6 rack screws with plastic washers to avoid damaging the rack ears.
3. Maintain at least 1U (44mm) of empty space above or below the amplifier for airflow.
4. The fan draws cool air from the front and expels hot air from the rear — ensure rear clearance of at least 100mm.
5. In dense rack configurations, consider a 1U ventilation panel above the amplifier.
6. Always route speaker cables away from signal cables to avoid interference.
7. Connect the chassis ground screw to the rack frame if hum is present.

### Standalone Operation

When not rack-mounted, place the amplifier on a stable, flat, ventilated surface. The rubber feet provide adequate clearance for bottom ventilation. Do not stack units directly on top of each other without rack mounting.

# 8. Wiring & Connection Guide

**STEREO / BRIDGE WIRING DIAGRAM**




**BRIDGE MODE:** Connect single speaker (min 8Ω) to CH A (+) and CH B (-) Speakon terminals. Set BRIDGE switch ON.

Figure 4 — Complete Stereo / Bridge Wiring Diagram

## Basic Stereo Connection (Recommended for most installations)

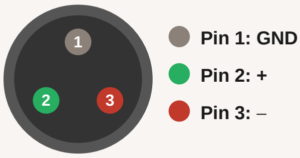
1. Power OFF both the mixer/source and the amplifier before connecting.
2. Connect the mixer's balanced XLR output (CH L) to the amplifier's XLR Input CH A.
3. Connect the mixer's balanced XLR output (CH R) to the amplifier's XLR Input CH B.
4. Connect a Speakon cable from Amplifier CH A Output to the Left speaker.
5. Connect a Speakon cable from Amplifier CH B Output to the Right speaker.
6. Set the Bridge/Parallel switch to STEREO.
7. Set gain knobs to minimum (0). Power on the amplifier, then slowly raise gain.

## Parallel Mode Connection

In Parallel mode, both CH A and CH B outputs carry the same signal (sourced from CH A input). This is useful for driving two speaker cabinets in parallel, or bi-amping with identical signal paths. Set the rear panel switch to PARALLEL. Connect source only to CH A XLR input.

## 9. XLR & Speakon Pinout Diagrams

**XLR INPUT PINOUT**




**SPEAKON OUTPUT PINOUT**

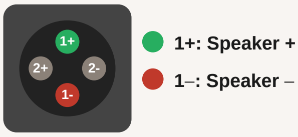



Figure 5 — XLR Input (left) and Speakon Output (right) Pinout Diagrams

### XLR Input Wiring Notes

- Pin 1 (GND/Shield): Connect to cable shield. Lift at source end if ground hum occurs.
- Pin 2 (+): Hot / positive / non-inverting signal.
- Pin 3 (–): Cold / negative / inverting signal.
- For unbalanced (RCA/TS) sources, use an active DI box or balanced adapter. Do not leave Pin 3 floating.
- Maximum input level before clipping: +20 dBu (at 32 dB gain setting).

### Speakon Output Wiring Notes

- Pins 1+ / 1–: Primary speaker signal. This is the standard connection for all single-channel loads.
- Pins 2+ / 2–: Used only for bi-amping with compatible 4-pole Speakon speaker cables.
- In Bridge mode: Connect 1+ (CH A) and 1– (CH B) to a single loudspeaker — minimum 8Ω.
- Always use Speakon cables with correctly rated conductor cross-section (min. 2.5mm² for long runs).
- Never connect or disconnect Speakon cables under signal — always mute/attenuate first.

## 10. Operating Modes

### Stereo Mode

Normal operation. Each channel (A and B) receives an independent input signal and drives an independent speaker load. Minimum speaker impedance: 4Ω per channel.

### Parallel Mode

CH A input is routed to both output channels simultaneously. Use when identical signal is needed for two speaker cabinets. Minimum impedance: 4Ω per channel (8Ω minimum if wiring speakers in parallel from one output).

### Bridge Mono Mode

Both channels are combined in anti-phase to produce a single high-power mono output. Connect speaker between CH A (+) and CH B (–) Speakon terminals. Minimum load: 8Ω. Do NOT use 4Ω loads in bridge mode — risk of damage.

| Mode | Min. Load | Max Power (TIP 2402) | Input Used |
|---|---|---|---|
| Stereo | 4Ω / ch | 4000W × 2 @ 4Ω | CH A + CH B |
| Parallel | 4Ω / ch | 4000W × 2 @ 4Ω | CH A only |
| Bridge Mono | 8Ω | 8000W @ 8Ω | CH A + CH B |

## 11. Gain & Sensitivity Setup

Correct gain structure is critical for maximum headroom and minimum noise. The TIP Series input sensitivity switchable between 0.775V (–2.2 dBu), 1.44V (+3.2 dBu), and 32 dB fixed gain. Follow this procedure:

1. Set all amplifier gain controls to minimum (fully counter-clockwise, position 0).
2. Set your mixer/source to its normal operating level (typically –10 dBu to +4 dBu peaks).
3. Set the Input Sensitivity switch to match your source: 0.775V for consumer/semi-pro, 1.44V for professional +4 dBu gear.
4. Play programme material at the loudest expected level.
5. Slowly raise the amplifier gain until the Signal LED illuminates consistently and the Clip LED flickers occasionally but not constantly.
6. Reduce gain slightly until clipping is rare. This is the optimal gain setting.
7. Document the gain knob position for future reference.

> ***NOTE:*** *Always set gain at the amplifier to match the source. Never rely solely on high gain at the amplifier with low source level — this raises the noise floor.*

# 12. Protection Systems

The TIP Series includes multi-stage automatic protection to safeguard both the amplifier and connected loudspeakers:

### Soft-Start / Inrush Control

On power-up, the unit delays full output for approximately 3 seconds, preventing transient pops that could damage tweeters.

### DC Offset Protection

Detects any DC voltage on the output (indicating an internal fault) and immediately mutes the output. Protect LED illuminates red.

### Short-Circuit Protection

Detects a shorted speaker cable or connector and mutes the affected channel. Inspect wiring before restarting.

### Over-Current Protection

If output current exceeds safe limits (below minimum impedance), the limiter activates. Check speaker impedance ratings.

### Thermal Protection (Over-Temperature)

The variable-speed fan increases speed with temperature. If temperature still exceeds the threshold, the unit reduces gain, then mutes to prevent damage. Allow to cool with mains power ON (fan continues to run).

### Clip Limiting

An analogue clip limiter reduces input level automatically when the output approaches clipping, providing transparent dynamic protection.

### RFI / EMI Filtering

Mains input includes filtering to reject radio-frequency and electromagnetic interference.

# 13. Troubleshooting

### No output / No power LED

→ Check AC mains connection and power switch.
→ Verify fuse in IEC inlet (replace with same type/rating only).
→ Check Power LED — if not lit, the unit may not be receiving power.

### Protect LED on / No audio output

→ Allow unit to cool for 10 minutes with power on.
→ Check speaker cables for short circuits.
→ Ensure speaker impedance meets minimum requirements (4Ω stereo, 8Ω bridge).
→ If Protect LED remains on after restart, contact authorised service.

### Hum or noise in output

→ Check all XLR cable shield connections (Pin 1).
→ Try lifting ground on source equipment.
→ Connect chassis ground screw to rack frame.
→ Ensure signal and power cables are routed separately.

### Clip LED always on

→ Reduce source level or amplifier gain.
→ Check if input sensitivity is set appropriately.

→ Ensure the input signal does not exceed +20 dBu.

### Distorted sound

→ Reduce gain to eliminate clipping.

→ Check speaker cables for intermittent shorts.

→ Verify speaker impedance — a 2Ω load in stereo mode will cause distortion.

### Unit runs very hot / Fan very loud

→ Verify adequate rack ventilation (min. 1U above and below).

→ Check that rear exhaust is not obstructed.

→ Consider reducing output level for thermally constrained installations.

## 14. Maintenance & Care

- Clean the exterior with a dry or slightly damp lint-free cloth. Never use solvents, alcohol, or abrasive cleaners.
- Periodically (every 6 months in dusty environments) use compressed air to clear dust from ventilation slots — with power disconnected.
- Inspect power and signal cable connectors for corrosion or physical damage before each use.
- Tighten rack mounting screws if vibration is present during operation.
- Do not attempt to lubricate or service the internal fan — this must be done by authorised service personnel.
- Store in the original packaging or a protective case when transporting.
- If the unit has not been used for more than 6 months, power it on briefly before use to re-form electrolytic capacitors.

## 15. Warranty & Support

Louis Martin Audio warrants this product against defects in materials and workmanship for a period of **2 years** from the date of original purchase. This warranty is valid for the original purchaser only and is non-transferable.

### Warranty Exclusions:

- Damage resulting from misuse, accidents, or modifications not authorised by Louis Martin Audio.
- Damage caused by operation outside specified voltage or impedance ranges.
- Cosmetic damage (scratches, dents) that does not affect functionality.
- Equipment used in environments not compliant with the safety instructions in Section 1.

### Contact & Support:

| | |
|---|---|
| **Website** | www.louismartinaudio.it |
| **Email** | contact@louismartinaudio.it |
| **Service** | Authorised service centres listed on the website |